Coding and Solution

AI/ML and Generative AI Track • Project Documentation

Project	Trip Genius AI (VoyagePlan Travel Planner)	**Date**	26 October 2026
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Implementation summary

The solution is split into a React/Vite frontend and a Python/FastAPI backend. The frontend manages the planning flow and calls the backend over HTTPS. The backend validates generation requests with Pydantic, calls Groq with server-side credentials and returns itinerary data. Supabase provides authentication and persistent trip records.

Core implementation flow

Step	Implementation responsibility
1	Collect and validate destination, dates, interests, budget and pace in the planner form.
2	Send the request to the FastAPI route POST /api/itineraries/generate with a loading state.
3	Validate request fields with backend Pydantic schemas before spending AI tokens.
4	Generate itinerary content with the configured Groq model, or return a clear error if the provider is unavailable.
5	Render the returned preferences and day-by-day itinerary in the frontend.
6	Use Supabase Auth for the user session and persist saved trips for the authenticated user.
7	Load saved trip values into the planner for editing and refresh the My Trips list after updates.

API and deployment

Field	Details
Generate itinerary	POST /api/itineraries/generate
Health check	GET /health — https://travel-itinerary-planner-1ysc.onrender.com/health
Backend (Render)	Root: backend | Build: pip install -r requirements.txt | Start: uvicorn app.main:app --host 0.0.0.0 --port $PORT | Python 3.11.9
Frontend (Vercel)	https://travel-itinerary-planner-0023-b9jioymcx-kumkum18.vercel.app
CORS	Exact Vercel origin allowed via ALLOWED_ORIGINS

Code evidence

Full source code, commit history and dependency files are available in the GitHub repository: https://github.com/thakurkumkum/travel-itinerary-planner. Verification of the solution is based on the production smoke tests described in the Project Testing folder.